



HSC Mathematics

Extension 2

Task 1 2016-2017

Time Allowed - 1 hour + 5minutes reading

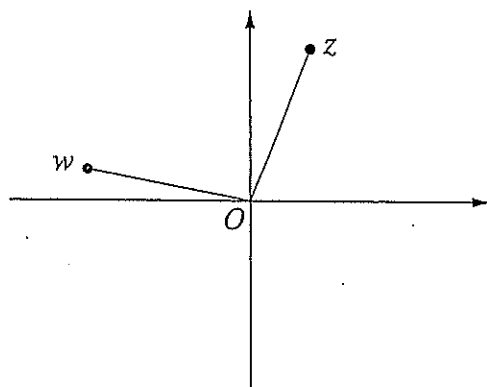
Instructions: Calculators may be used in any parts of the task. For 1 Mark Questions, the correct answer is sufficient to receive full marks. For Questions worth more than 1 Mark, necessary working MUST be shown to receive full marks.

Multiple Choice	/4
Question 5	/12
Question 6	/14
Question 7	/14
Total	/44

2016 - 2017 Extension 2 Assessment Task 1

Answer on the multiple choice answer sheet (1 mark each)

- 1 The Argand diagram shows the complex numbers z and w , where z lies in the first quadrant and w lies in the second quadrant.



Which complex number could lie in the 3rd quadrant?

- (A) $-w$ (C) $\bar{z}$
 (B) $2iz$ (D) $w - z$
- 2 Given $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, which expression is equal to $(\bar{z})^{-1}$?

- (A) $\frac{1}{2}\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$ (C) $\frac{1}{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$
 (B) $2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$ (D) $2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

- 3 The complex number z satisfies $|z - 1| = 1$.

What is the greatest distance that z can be from the point i on the Argand diagram?

- (A) 1 (C) $2\sqrt{2}$
 (B) $\sqrt{5}$ (D) $\sqrt{2} + 1$
- 4 The cartesian equation of the locus specified by $|z|^2 = z + \bar{z}$ is
- A. $(x - 1)^2 + y^2 = 1$ B. $(x + 1)^2 + y^2 = 1$
 C. $x^2 + (y - 1)^2 = 1$ D. $x^2 + (y + 1)^2 = 1$

Question 5 (12 Marks) Begin a New Sheet of Paper

- | | | Marks |
|------|--|-------|
| a) | Given $A = 3 + 4i$ and $B = 1 - i$, express the following in the form $x + iy$ | |
| (i) | AB | 1 |
| (ii) | $\frac{A}{iB}$ | 1 |
| | | |
| b) | Let $\alpha = -\sqrt{3} + i$ | |
| (i) | Find the exact value of $ \alpha $ and $\arg \alpha$ | 2 |
| (ii) | Find the exact value of α^5 in the form $a + ib$ where a and b are real. | 2 |
| | | |
| c) | Sketch the locus of the point representing the complex number z on the Argand diagram and find its Cartesian equation: | |
| (i) | $ z + i = z - 1 $ | 2 |
| (ii) | $\operatorname{Arg} \left(\frac{z+1}{z-i} \right) = \frac{\pi}{2}$ | 3 |
| | | |
| (d) | Multiplying a non-zero complex number by $\frac{1-i}{1+i}$ results in a rotation about the origin on an Argand diagram. | |
| | What is the rotation? | 1 |

- (a) Let $z_1 = \cos \theta + i \sin \theta$ and $z_2 = \cos \phi + i \sin \phi$, where θ and ϕ are real.

Show that: (i) $\frac{1}{z_1} = \cos \theta - i \sin \theta$

1

(ii) $z_1 z_2 = \cos(\theta + \phi) + i \sin(\theta + \phi)$

1

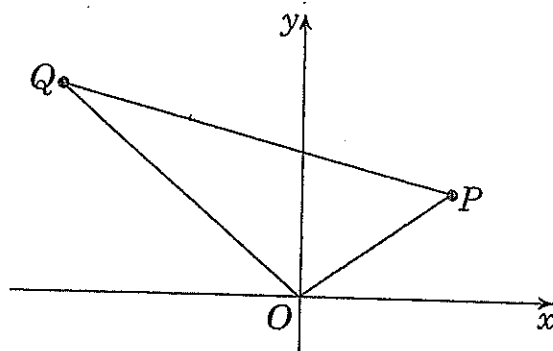
- (b) (i) Find all pairs of integers x and y such that $(x + iy)^2 = -3 - 4i$.

2

(ii) Using (i), or otherwise, solve the quadratic equation $z^2 - 3z + (3 + i) = 0$.

2

c)



The diagram shows a complex plane with origin O . The points P and Q represent arbitrary non-zero complex numbers z_1 and z_2 respectively.

- (i) Use the diagram to show that

$$|z_1 - z_2| \leq |z_1| + |z_2|.$$

2

- (ii) $|z_1 - z_2| \geq ||z_1| - |z_2||$.

2

- (iii) Construct the point R representing $z_1 + z_2$.
What type of quadrilateral is $OPRQ$?

2

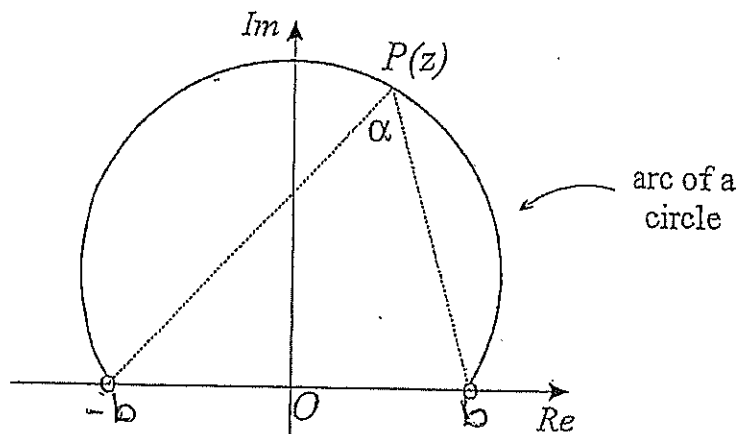
- (iv) If $|z_1 - z_2| = |z_1 + z_2|$, what can be said
about the complex number $\frac{z_1}{z_2}$?

2

Question 7 (14 Marks) Begin a New Sheet of Paper

Marks

(a)



1

In the diagram above, the locus of the point P representing the complex number z is graphed. Write down a possible equation in terms of z , b and α for the locus of P. Note that constants b and α are real.

b) (i) Use De Moivre's Theorem, to find expressions for $\cos 3\theta$ and $\sin 3\theta$

3

(ii) Hence prove that $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$

2

c) It is given that $z^5 = 1$ where $z \neq 1$

(i) Show that $z^2 + z + 1 + z^{-1} + z^{-2} = 0$

2

(ii) Show that $z + z^{-1} = 2 \cos \frac{2k\pi}{5}$ $k = 1, 2, 3, 4$.

2

(iii) By letting $x = z + z^{-1}$ reduce the equation in (i) above to a quadratic equation in x .

2

(iv) Hence deduce that $\cos \frac{\pi}{5} \cdot \cos \frac{2\pi}{5} = \frac{1}{4}$

2

1) D 2) C

3) D 4) A

Q5.

a) $A = 3+4i$ $B = 1-i$

i) $AB = (3+4i)(1-i)$
 $= 3 - 3i + 4i + 4$
 $= 7+i$

ii) $\frac{A}{iB} = \frac{3+4i}{1+i} \times \frac{1-i}{1-i}$

$= \frac{7+i}{2}$

b) $\alpha = -\sqrt{3}+i$

i) $|\alpha| = \sqrt{3+1}$
 $= 2$
 $\tan \beta = \frac{-1}{\sqrt{3}}$
 $\therefore \arg \alpha = \frac{5\pi}{6}$

ii) $\alpha = 2 \operatorname{cis} \frac{5\pi}{6}$

$\alpha^5 = 2^5 \left(\operatorname{cis} \frac{25\pi}{6} \right)$
 $= 32 \operatorname{cis} \frac{\pi}{6}$
 $= 32 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$
 $= 32 \left(\frac{\sqrt{3}}{2} + i \cdot \frac{1}{2} \right)$
 $= 16\sqrt{3} + 16i$

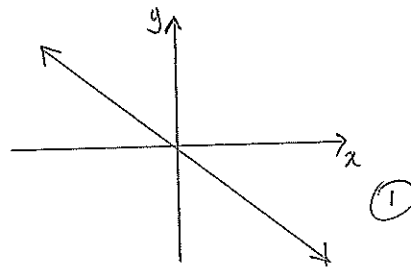
c) i) $|z+i| = |z-1|$

$|x+(y+1)i| = |(x-1)+iy|$

$x^2 + y^2 + 2y + 1 = x^2 - 2x + 1 + y^2$

$2y = -2x$

$\therefore y = -x$ (1)



ii) $\operatorname{Arg} \left(\frac{z+1}{z-i} \right) = \frac{\pi}{2} \rightarrow \text{undefined for } -1 \text{ and } i$

i.e. $\operatorname{Arg}(z+1) - \operatorname{Arg}(z-i) = \frac{\pi}{2}$

let $\operatorname{Arg}(z+1) = \alpha$ and $\operatorname{Arg}(z-i) = \beta$

$\therefore \alpha - \beta = \frac{\pi}{2}$

$\alpha = \frac{\pi}{2} + \beta$

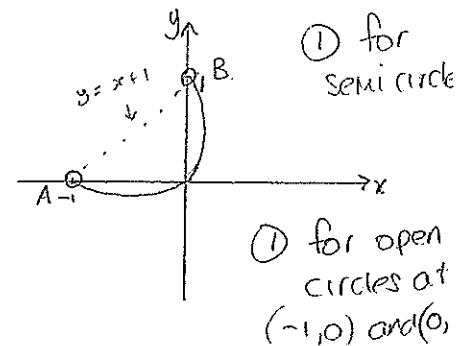
i.e. $\alpha > \beta$

$\angle AZB = \frac{\pi}{2}$ $\therefore AB$ is diameter of circle, centre $(-\frac{1}{2}, \frac{1}{2})$
 radius $\frac{\sqrt{2}}{2}$ (1) for eqn

i.e. $\left(x + \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{2}, y < x+1$

d) $\frac{1-i}{1+i} \cdot \frac{1-i}{1-i} = \frac{1-2i-1}{1+1}$
 $= -i$

$\therefore$ clockwise by $\frac{\pi}{2}$ (1)
 or $-\frac{\pi}{2}$



Q6

a) $z_1 = \cos \theta, z_2 = \cos \phi$

i) $\frac{1}{z_1} = (z_1)^{-1}$
 $= (\cos \theta + i \sin \theta)^{-1}$
 $= \cos(-\theta) + i \sin(-\theta)$
 (by de Moivre)
 $= \cos \theta - i \sin \theta$
 $= \text{RHS} //$

ii) $z_1 z_2 = \cos(\theta + \phi) + i \sin(\theta + \phi)$
 LHS $= (\cos \theta + i \sin \theta)(\cos \phi + i \sin \phi)$
 $= \cos \theta \cos \phi + i \sin \theta \cos \phi + i \sin \theta \sin \phi - \sin \theta \sin \phi$
 $= (\cos \theta \cos \phi - \sin \theta \sin \phi) + i(\sin \theta \cos \phi + \sin \phi \cos \theta)$
 $= \cos(\theta + \phi) + i \sin(\theta + \phi)$

b) $(x + iy)^2 = -3 - 4i$

$x^2 - y^2 = -3$

$2xy = -4 \Rightarrow y = \frac{-2}{x}$

$\therefore x^2 - \frac{4}{x^2} = -3$

$x^4 + 3x^2 - 4 = 0$

$(x^2 + 4)(x^2 - 1) = 0$

$x = \pm 1, x, y \in \mathbb{R}$

$x = 1 \quad | \quad x = -1$

$y = -2 \quad | \quad y = 2$

ii) $z^2 - 3z + (3 + i) = 0$

$z = \frac{3 \pm \sqrt{9 - 4(3 + i)}}{2}$

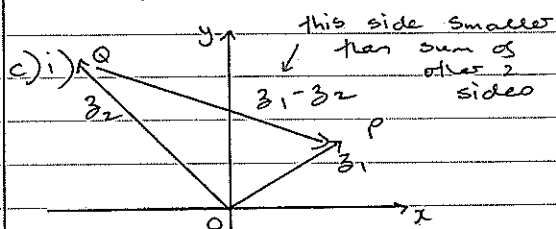
$= \frac{3 \pm \sqrt{9 - 12 - 4i}}{2}$

$= \frac{3 \pm \sqrt{-3 - 4i}}{2}$

$= \frac{3 \pm (1 - 2i)}{2}$

$= \frac{3 + 1 - 2i}{2}, \frac{3 - 1 + 2i}{2}$

$= 2 - i, 1 + i$



$|z_1 - z_2| \leq |z_1| + |z_2|$

$|z_1 - z_2|$ is the side $\overrightarrow{QP}$ (ie length)

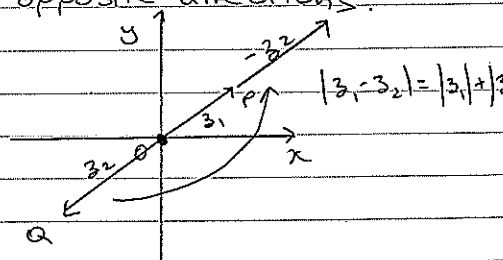
$|z_1|$ is the side $\overrightarrow{OP}$ (length)

$|z_2|$ is the side $\overrightarrow{OQ}$ (length)

$\therefore \overrightarrow{QP}$ has length less than the sum of the other 2 sides in the triangle.

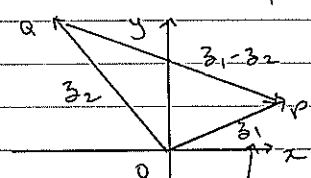
Equality when $\overrightarrow{OP} + \overrightarrow{OQ}$ are parallel

in opposite directions.



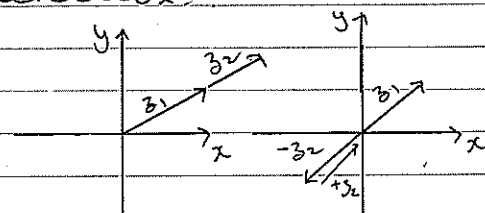
ii) $|z_1 - z_2| \geq ||z_1| - |z_2||$

ie $|z_1 - z_2| + |z_2| \geq |z_1|$

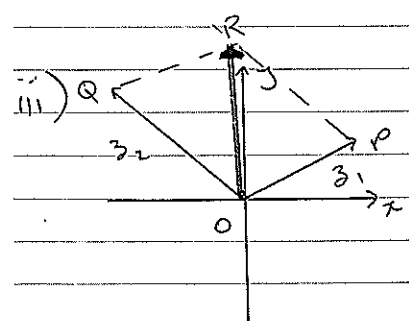


$|z_1|$ is smaller than the sum of the other 2 sides.

Equality when z_1 & z_2 are parallel & in the same direction



$|z_1 - z_2| + |z_2| = |z_1|$



$\overrightarrow{OR}$ represents $z_1 + z_2$

$\therefore OPRQ$ is a parallelogram

iv) If $|z_1 - z_2| = |z_1 + z_2|$
 diagonals equal
 $\therefore OPRQ$ is rectangle.

$\arg z_1 - \arg z_2 = \pm \frac{\pi}{2}$

$\therefore \frac{z_1}{z_2} = \pm ki \rightarrow$ Purely Imaginary

$$7) a) \arg(z-b) + \alpha = \arg(z-b)$$

$$\alpha = \arg(z-b) - \arg(z+b) \\ = \arg\left(\frac{z-b}{z+b}\right)$$

$$b) i) (\cos\theta + i\sin\theta)^3 = (\operatorname{cis}\theta)^3 \quad \text{by de Moivre's theorem} \\ = \operatorname{cis}3\theta \\ = \cos3\theta + i\sin3\theta$$

also

$$(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3\cos^2\theta\sin\theta + 3i\cos\theta\sin^2\theta + i^3\sin^3\theta \\ = \cos^3\theta - 3\cos\theta\sin^2\theta + i(3\cos^2\theta\sin\theta - \sin^3\theta)$$

equating real parts

$$\cos3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$$

equating imaginary parts

$$\sin3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta$$

$$ii) \tan3\theta = \frac{\sin3\theta}{\cos3\theta} \\ = \frac{3\cos^2\theta\sin\theta - \sin^3\theta}{\cos^3\theta - 3\cos\theta\sin^2\theta}$$

$\div$ top & bottom by $\cos^3\theta$

$$= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

$$c) i) z^5 = 1 \\ z^5 - 1 = 0$$

$$(z-1)(z^4 + z^3 + z^2 + z + 1) = 0$$

since $z \neq 1 \quad \div$ both sides by $z-1$

$$z^4 + z^3 + z^2 + z + 1 = 0$$

since $z \neq 0 \quad \div$ both sides by z^2

$$z^2 + z + 1 + z^{-1} + z^{-2} = 0$$

$$ii) z^5 = 1$$

$$z^5 = \operatorname{cis}(0 + 2k\pi) \quad \text{for } k \in \mathbb{Z}$$

$$z = \operatorname{cis} \frac{2k\pi}{5} \quad \text{but } k \neq 0, 5, 10, \dots$$

since $z \neq 1$

$$z^{-1} = \left(\operatorname{cis} \frac{2k\pi}{5}\right)^{-1}$$

$$= \operatorname{cis}\left(-\frac{2k\pi}{5}\right)$$

$$z + z^{-1} = \operatorname{cis}\left(\frac{2k\pi}{5}\right) + \operatorname{cis}\left(-\frac{2k\pi}{5}\right) \\ = 2\cos \frac{2k\pi}{5}$$

$$iii) z^2 + z + 1 + z^{-1} + z^{-2} = 0$$

$$z^2 + z + z^{-2} + z + z^{-1} + 1 - 2 = 0$$

$$(z + z^{-1})^2 + z + z^{-1} - 1 = 0$$

$$x^2 + x - 1 = 0$$

$$\text{iv)} \quad x^2 + x - 1 = 0$$

let the roots be α, β

$$\Delta = (1)^2 - 4 \times 1 \times -1$$

$$= 1 + 4$$

$$= 5$$

$$\therefore \alpha, \beta \in \mathbb{R} \text{ \& } \alpha \neq \beta$$

$$\cancel{\text{Note}} \quad x = z + z^{-1} = 2 \cos \frac{2k\pi}{5} \text{ for } k=1, 2, 3, 4$$

$$\text{Note that for } k=4 \quad x = 2 \cos \frac{8\pi}{5}$$

$$= 2 \cos \frac{2\pi}{5}$$

which is the same as $k=1$

$$\text{for } k=3 \quad x = 2 \cos \frac{6\pi}{5}$$

$$= 2 \cos \frac{4\pi}{5}$$

which is the same as $k=2$

$$\therefore \alpha = 2 \cos \frac{2\pi}{5} \text{ \& } \beta = 2 \cos \frac{4\pi}{5} \text{ or vice versa.}$$

$$\text{by product of roots } \alpha\beta = \frac{-1}{1}$$

$$2 \cos \frac{2\pi}{5} \times 2 \cos \frac{4\pi}{5} = -1$$

$$4 \cos \frac{2\pi}{5} \times -\cos \frac{\pi}{5} = -1$$

$$\cos \frac{\pi}{5} \cos \frac{2\pi}{5} = \frac{1}{4}$$